

Enzyme-Unified: Learning Unified and Holistic Enzyme Function Representations via Hybrid Interaction Modeling

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Abstract

Accurately predicting diverse functional properties of enzymes remains a central challenge in biotechnology. Many existing machine learning methods are limited by two fundamental issues: they treat each property as an independent task with simplistic feature fusion, thereby ignoring important relationships among properties, and they are commonly evaluated on homology-biased datasets, leading to overly optimistic performance estimates. To address these limitations, we present ENZYME-UNIFIED, a unified learning framework based on a single, expressive architecture. The framework is instantiated to model five essential enzymatic properties: turnover number, Michaelis constant, catalytic efficiency, optimal temperature, and optimal pH. At the heart of our approach is a novel hybrid interaction model that dynamically combines fine-grained local interactions captured by cross-attention with global feature representations through a trainable gating mechanism, enabling comprehensive modeling of enzyme function. For rigorous assessment, we construct three large-scale benchmark datasets with enforced sequence dissimilarity. Experimental results demonstrate that ENZYME-UNIFIED consistently achieves state-of-the-art performance and substantially surpasses existing approaches on these carefully curated benchmarks, highlighting the effectiveness of its hybrid architecture. The code and datasets will be released upon acceptance.

1. Introduction

Enzymes are nature’s catalysts, driving biochemical reactions that are fundamental to life sciences and a cornerstone

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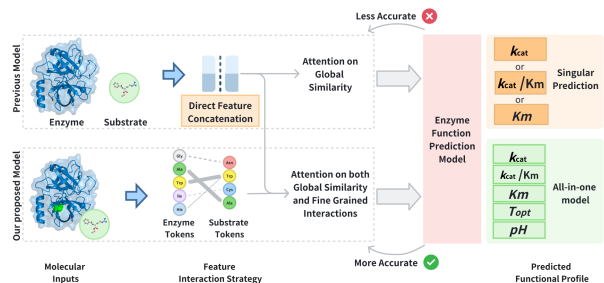


Figure 1. Conceptual comparison of our framework with previous models. **(Top)** Prior methods, based on coarse global features, yield fragmented singular predictions. **(Bottom)** By capturing fine-grained interactions, our model provides a holistic, *all-in-one* prediction of the enzyme’s entire functional profile, including kinetics and environmental optima.

of modern industry, from pharmaceuticals to biofuel production. (Wu et al., 2021; Zhang et al., 2015; Guldhe et al., 2015) Harnessing their full potential, however, requires tailoring these biocatalysts to the specific demands of industrial applications, a task for which natural enzymes are rarely optimized. A central goal of enzyme engineering is to design or discover novel catalysts with desired functional properties. Key metrics such as catalytic rate (k_{cat}), the Michaelis constant (K_m), catalytic efficiency (k_{cat}/K_m), optimal temperature (T_{opt}), and optimal pH dictate an enzyme’s utility. However, the experimental characterization of these properties is often a laborious, costly, and low-throughput process. In recent years, deep learning has emerged as a powerful paradigm to predict these properties from sequence and substrate information, promising to accelerate computational enzyme design. This progress is largely driven by advances in protein language models, which learn deep, context-aware representations of protein sequences from vast unlabeled databases (Lin et al., 2023; Elnaggar et al., 2021). For instance, models like UniKP have shown success in predicting key kinetic parameters from sequence and substrate information (Yu et al., 2023), while others, such as PRIME, have focused on environmental properties like thermostability (Jiang et al., 2024), promising to replace slow experimental cycles with rapid in silico screening.

Despite promising progress, current approaches are plagued by two fundamental limitations. First, they are fragmented into a single-task research paradigm, where models are

designed to predict a single property in isolation. This disregards the intricate biophysical interplay between an enzyme’s catalytic activity and the environmental conditions in which it functions. Moreover, this fragmentation often leads to models being evaluated on improperly split datasets that fail to account for sequence homology. Consequently, reported performance can be artificially inflated, masking poor generalization to novel enzymes, which is a critical failure for real-world discovery applications.

This fragmentation significantly curtails the practical utility of predictive models. A highly active enzyme (high k_{cat}/K_m) is of little industrial value if its optimal conditions (T_{opt} , pH) are impractically narrow or extreme. To guide meaningful experiments, scientists require a holistic view of an enzyme’s performance profile. Therefore, a unified framework that can co-predict multiple, interdependent properties, a challenge well-suited for a multi-task learning paradigm, which has proven effective for improving generalization and data efficiency in modern deep learning (Zhang & Yang, 2021), is not merely an incremental improvement, but a necessary step towards building truly applicable computational tools.

To bridge this gap, we introduce ENZYME-UNIFIED, a multi-task learning framework designed to provide a comprehensive and robust profile of enzyme function by simultaneously predicting catalytic rate, the Michaelis constant, catalytic efficiency, optimal temperature, and optimal pH. At the architectural level, we address the reliance on simple feature concatenation, another key weakness of prior work. We propose a novel hybrid interaction model that dynamically integrates two distinct paradigms: a global feature concatenation network and a token-level cross-attention network designed to capture fine-grained molecular interactions. This latter approach has shown significant promise for explicitly modeling interactions between distinct molecular entities, such as proteins and small molecules (Huang et al., 2021). Crucially, our model adaptively balances these two pathways through a single, end-to-end trainable gating parameter, allowing it to learn the optimal feature abstraction strategy for a given enzyme-substrate pair. Our primary contributions are summarized as follows:

- We propose ENZYME-UNIFIED, a unified framework for holistic enzyme property prediction, powered by a novel HYBRID INTERACTION MODEL that adaptively fuses global and local interaction features for more powerful and flexible representations.
- We construct three new large-scale, rigorously partitioned datasets for multi-property prediction, which are carefully split by sequence similarity to ensure unbiased evaluation and promote model generalization.
- Our framework achieves state-of-the-art performance

on the public, unbiased CataPro benchmark and significantly outperforms specialized single-task baselines across all properties on our new datasets.

2. Related Work

From Single-Task to Holistic Prediction. Deep learning can handle multiple tasks simultaneously (Zhang & Yang, 2021; Zeng et al., 2025). Early machine learning models laid a foundation by predicting isolated enzyme properties, such as catalytic rate (k_{cat}) or substrate affinity (K_m) (Li et al., 2022; Kroll et al., 2021), while a parallel line of research targeted thermostability (T_m). These single-task approaches, however, fail to capture intricate biophysical trade-offs. Recognizing this, recent models such as UniKP advanced towards jointly predicting kinetic parameters (k_{cat} , K_m) (Yu et al., 2023). Still, a critical gap remains: these models treat environmental factors like pH and temperature as conditional inputs rather than properties to be predicted. This prevents the modeling of a holistic functional profile, including an enzyme’s apparent optimal conditions (T_{opt} and optimal pH)¹. In contrast, our work, ENZYME-UNIFIED, is to our knowledge the first framework to treat an enzyme’s kinetics and its optimal conditions as coequal targets, thus providing a truly holistic functional profile.

The Imperative for Unbiased Evaluation. The field has increasingly recognized that model performance on randomly split benchmarks is often an illusion, inflated by “data leakage” from sequence homology (Rao et al., 2019; Jumper et al., 2021). Recent works like CataPro (Wang et al., 2025) and CatPred (Boorla & Maranas, 2025) have established a new, more rigorous standard by introducing benchmarks with strict, non-homologous data splits and a focus on out-of-distribution generalization. Our work fully adheres to this philosophy and the demand for rigorous, unbiased evaluation. We validate our model’s performance on the public CataPro benchmark and further contribute to this effort by developing and releasing three new, large-scale datasets partitioned by a strict sequence similarity threshold to facilitate robust, multi-property model evaluation.

Architectures for Enzyme-Substrate Interaction. At the architectural level, a common strategy involves encoding enzyme and substrate using pre-trained models like ProtT5 (Elnaggar et al., 2021) and SMILES transformers (Kreutter et al., 2021), respectively. However, the fusion of these representations is often accomplished via simple concatenation (Yu et al., 2023; Wang et al., 2025; Boorla & Maranas, 2025), which may not adequately capture fine-grained,

¹Importantly, in our setting, the labels for these optimal conditions are drawn from heterogeneous assay measurements. Where multiple optima are reported for the same enzyme-substrate pair under varying experimental setups, we resolve them by taking the median within that pair (but never across different substrates).

token-level interactions at the active site. We directly address the limitation for modeling enzyme-substrate interactions with a novel hybrid interaction model, which moves beyond static concatenation by adaptively fusing global features with detailed, token-level interactions learned via a trainable cross-attention mechanism. Beyond these directions, several recent works have explored enzyme-substrate modeling for different problem settings. For example, EnzymeCAGE (Liu et al., 2024) focuses on hierarchical multi-label prediction to improve multi-label enzyme function prediction. EZSpecificity (Cui et al., 2025) treats reactivity as a ranking problem to select likely substrates. Inspired by these works, we develop ENZYME-UNIFIED.

3. Method

Our methodology is structured into three key components. We begin in Section 3.1 by detailing the creation and rigorous, non-homologous partitioning of three new, large-scale datasets, which form the foundation for both training and unbiased evaluation. Next, in Section 3.2, we introduce the ENZYME-UNIFIED framework, a novel hybrid architecture designed to overcome the limitations of previous models by holistically predicting multiple enzyme properties. We describe its multi-modal input representations and its core hybrid interaction model. Finally, Section 3.3 outlines the objective function and training protocol used to optimize the framework for each predictive task.

3.1. Dataset Construction and Unbiased Partitioning for Multi-Task Learning

To rigorously train and evaluate our multi-task framework, we first established a suite of three new, large-scale datasets for predicting catalytic efficiency (k_{cat}/K_m), optimal temperature (T_{opt}), and optimal pH. Recognizing that simple random splits are insufficient for robust model evaluation due to sequence homology, which can lead to overly optimistic performance estimates (Rao et al., 2019), we implemented a meticulous, cluster-based partitioning strategy from the outset.

Our standardized protocol ensures a reproducible and unbiased partitioning for all datasets. The protocol employs MMseqs2 (Steinegger & Söding, 2017) to cluster all unique protein sequences based on a strict 40% sequence identity threshold, which guarantees that highly similar proteins are grouped into the same cluster (Rost, 1999). Subsequently, we distribute these entire clusters, rather than individual data points, into distinct and non-overlapping folds. This assignment is performed greedily: clusters are sorted by size and then iteratively allocated to the fold with the fewest currently assigned samples to maintain balance. Refer to Appendix E for more details. Applying this rigorous protocol, we construct and partition the following three datasets.

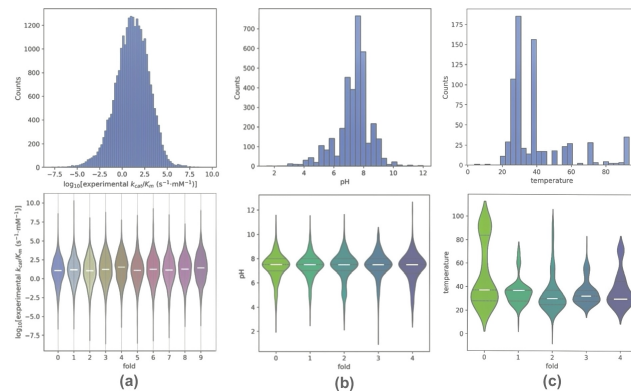


Figure 2. Statistical distribution of the curated datasets. The figure displays the overall distribution (top) and the per-fold distribution (bottom) for (a) catalytic efficiency (k_{cat}/K_m), (b) optimal pH, and (c) optimal temperature (T_{opt}). The consistency across folds demonstrates our balanced, cluster-based data partitioning.

The Catalytic Efficiency (k_{cat}/K_m) Dataset. The construction of this dataset involved a comprehensive, multi-stage pipeline. We began by aggregating all entries containing k_{cat} or K_m values from the BRENDA (Chang et al., 2021) and SABIO-RK (Wittig et al., 2012) databases. Of 30,995 unique substrate names, 17,610 could not be reliably mapped to a SMILES string and were excluded. The remaining data were cleaned, and for entries with multiple annotations, we retained the maximum k_{cat} and minimum K_m to capture peak performance. After integrating all sources and performing a final deduplication, our dataset consists of 26,361 high-confidence samples. The log10-transformed values are approximately normally distributed, as is shown in Figure 2a. Following our partitioning protocol, the sequences were grouped into 3,329 clusters and distributed into 10 balanced folds for cross-validation, with consistent value distributions across all folds.

The Optimal pH Dataset. Our curation for this dataset began with an initial collection of 6,521 raw entries. The data was subjected to a multi-stage filtering pipeline, removing entries without a valid substrate SMILES string or enzyme sequence. We retained the median value for enzymes with multiple reported optimal pH values. This pipeline yielded our final dataset of 3,435 high-quality samples exhibiting bimodal distribution, as shown in Figure 2b. The sequences were grouped into 2,238 clusters and partitioned into 5 balanced folds maintaining a consistent pH distribution.

The Apparent Temperature Optimum (T_{opt} , °C) Dataset. This dataset was constructed from an initial pool of 2,233 entries. We applied a stringent cleaning protocol, discarding records lacking essential information and taking the median value for enzymes with multiple reported optima. The resulting 706 unique entries represent the temperature optimum, which reflects the balance between catalytic activity

and thermal stability under the specific assay timescale and experimental matrix, rather than a purely intrinsic property of the enzyme. As shown in Figure 2c, the values are concentrated around 30–40°C. The sequences were clustered into 251 groups and partitioned into 5 folds, with consistent value distributions across folds to ensure a fair evaluation.

This unified methodology of curation and cluster-based partitioning ensures that any given protein sequence and its close homologs are strictly confined to a single fold. Consequently, during cross-validation, the model is always evaluated on enzymes that are significantly different from those seen during training. This forces the model to learn generalizable sequence-function relationships and provides a robust foundation for all experiments reported in this work.

3.2. The Enzyme-Unified Framework

Current deep learning approaches for enzyme function are limited by a single-task paradigm that ignores the **intrinsic biophysical coupling between kinetics and environmental optima**, and by an oversimplified concatenation of global features that misses fine-grained interactions. To overcome these challenges, we propose ENZYME-UNIFIED, a framework built on a novel hybrid architecture. It moves beyond simple concatenation by explicitly modeling and dynamically balancing both global and fine-grained, token-level interactions to learn a more comprehensive representation. This allows our framework to provide a truly holistic functional profile by predicting five distinct properties together: k_{cat} , K_m , k_{cat}/K_m , T_{opt} , and pH.

3.2.1. MULTI-MODAL INPUT REPRESENTATION

The framework takes a multi-modal input, generating representations for both the enzyme and substrate from pre-trained language models and chemical fingerprints.

Enzyme Representation: We use the ProtT5-XL-U50 (Elnaggar et al., 2021) encoder to generate token-level embeddings from the amino acid sequence, denoted as $\mathbf{E} \in \mathbb{R}^{L \times 1024}$. We also use a pre-computed, mean-pooled global embedding, $\mathbf{h}_E \in \mathbb{R}^{1024}$.

Substrate Representation: The substrate’s SMILES string is encoded using the MolT5 (Edwards et al., 2022) encoder, yielding token-level embeddings $\mathbf{S} \in \mathbb{R}^{M \times 768}$. A global representation is formed by concatenating the pre-computed, mean-pooled MolT5 embedding ($\mathbf{h}_S \in \mathbb{R}^{768}$) with a 167-dimensional MACCS chemical fingerprint ($\mathbf{h}_{maccs} \in \mathbb{R}^{167}$) (Durant et al., 2002).

Auxiliary Physicochemical Features: As an optional input, we incorporate a 22-dimensional vector of pre-computed physicochemical properties of the enzyme, denoted as $\mathbf{h}_{phys} \in \mathbb{R}^{22}$. These descriptors, including amino acid composition, molecular weight, and isoelectric point, were cal-

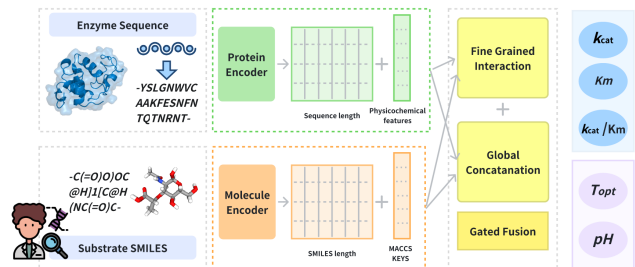


Figure 3. Architecture of the Enzyme-Unified hybrid framework. The framework processes encoded enzyme and substrate inputs through parallel pathways: a fine-grained interaction path to capture localized interactions, a global concatenation path for pooled features, and a gated fusion mechanism to predict the complete profile of kinetics and environmental optima dynamically.

culated from the primary sequence using the ProtParam module of the Biopython library (Cock et al., 2009).

3.2.2. THE HYBRID INTERACTION ARCHITECTURE

Our key architectural innovation is a hybrid model comprising two distinct sub-models, or pathways, that capture different interaction philosophies. The final prediction is a learned interpolation of outputs from these two pathways.

The Fine-Grained Interaction Pathway: This pathway is designed to model localized, molecular-level interactions through a stack of N bidirectional cross-attention layers. Let $\mathbf{E}^{(l)}$ and $\mathbf{S}^{(l)}$ be the token-level representations of the enzyme and substrate at layer l , respectively. With initial inputs $\mathbf{E}^{(0)}$ and $\mathbf{S}^{(0)}$ from projected embeddings of T5 encoders, output of each layer is defined as:

$$[\mathbf{E}^{(l)}, \mathbf{S}^{(l)}] = \text{BiCROSSATTN}^{(l)}(\mathbf{E}^{(l-1)}, \mathbf{S}^{(l-1)}), \quad (1)$$

where the BiCrossAttn block internally computes parallel multi-head attention updates (enzyme-attending-to-substrate and vice-versa), followed by feed-forward networks. After the final layer, the contextualized representations $\mathbf{E}^{(N)}$ and $\mathbf{S}^{(N)}$ are pooled and combined with the projected MACCS fingerprint embedding $\mathbf{h}'_{maccs}$ to form a final feature vector, $\mathbf{z}_{attn}$. A dedicated MLP then generates the prediction:

$$\begin{aligned} \mathbf{z}_{ATTN} &= [\text{MEANPOOL}(\mathbf{E}^{(N)}); \text{MEANPOOL}(\mathbf{S}^{(N)}); \mathbf{h}'_{MACCS}] \\ \hat{\mathbf{y}}_{ATTN} &= \text{MLP}_{ATTN}(\mathbf{z}_{ATTN}). \end{aligned} \quad (2)$$

This hierarchical process allows the model to build progressively more sophisticated representations of the enzyme-substrate complex.

The Global Concatenation Pathway: This pathway reflects the traditional modeling approach by operating exclusively on global feature vectors. It takes the mean-pooled

enzyme embedding ($\mathbf{h}_E$), the mean-pooled substrate embedding ($\mathbf{h}_S$), and the substrate’s MACCS fingerprint ($\mathbf{h}_{maccs}$). The language model embeddings are first independently normalized using BatchNorm1d (Ioffe & Szegedy, 2015) to stabilize distributions. These normalized features are then concatenated to form a single feature vector, $\mathbf{z}_{cat}$, which is processed by a dedicated MLP decoder to yield a prediction:

$$\mathcal{L}_t = \frac{1}{N_t} \sum_{i=1}^N \mathbf{1}_{\{y_i^{(t)} \text{ avail.}\}} (\hat{y}_i^{(t)} - T_t(y_i^{(t)}))^2, \quad (3)$$

for $t \in \{k_{cat}, K_m, k_{cat}/K_m, \text{pH}, T_{opt}\}$,

This pathway serves as a robust baseline, capturing the overall context of the enzyme-substrate pair.

Gated Prediction Fusion: The final prediction of the hybrid model, $\hat{y}$, is a learned interpolation of the outputs from the two pathways. The balance is controlled by a single, globally trainable scalar parameter, α , allowing the optimization process to find an optimal, task-specific balance between the two modeling strategies. The final prediction is computed as:

$$\hat{y} = \sigma(\alpha) \cdot \hat{y}_{attn} + (1 - \sigma(\alpha)) \cdot \hat{y}_{concat}, \quad (4)$$

where σ is the sigmoid function, this formulation allows the optimization process to find an optimal balance between the two modeling strategies for each predictive task.

3.3. Objective Function and Training Protocol

Our unified architectural framework is designed to support both specialized training (one model per task) and true multi-task training (shared encoder with multiple prediction heads). For the benchmark results reported in this paper, we adopt the specialized training protocol: A separate instance of the unified architecture is trained for each of the five target properties. This avoids the complexities of multi-task loss balancing and allows each model to fully optimize its regression objective.

Formally, each task is trained with a standard Mean Squared Error (MSE) loss. Because kinetic parameters span several orders of magnitude, we apply a task-specific transformation $T(y)$ before regression:

$$T(y) = \begin{cases} \log_{10}(y), & \text{if } y \in \{k_{cat}, K_m, \frac{k_{cat}}{K_m}\}, \\ y, & \text{otherwise (pH, } T_{opt}). \end{cases} \quad (5)$$

Given predictions $\hat{y}_i = \mathcal{F}_\theta(x_i)$, the single-task objective is:

$$\mathcal{L}_{\text{single}}(\theta) = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - T(y_i))^2. \quad (6)$$

The models are trained using AdamW (Loshchilov & Hutter, 2019), with early stopping based on validation RMSE (evaluated on the original scale) to prevent overfitting.

Although our main results use specialized training, our architecture naturally supports a true multi-task regime. In the multi-task setting, the five task-specific losses are optimized jointly, with gradients flowing through a shared encoder and task-specific heads. The multi-task objective is:

$$\mathcal{L}_t = \frac{1}{N_t} \sum_{i=1}^N \mathbf{1}_{\{y_i^{(t)} \text{ avail.}\}} (\hat{y}_i^{(t)} - T_t(y_i^{(t)}))^2, \quad (7)$$

for $t \in \{k_{cat}, K_m, k_{cat}/K_m, \text{pH}, T_{opt}\}$.

This formulation enables the encoder to capture shared biochemical representations across tasks, while allowing each head to model property-specific variations. Our expanded experiments demonstrate that the same unified architecture remains competitive or superior under both specialized and multi-task training protocols.C.1

4. Experiments

4.1. Datasets and Preprocessing

A robust and diverse collection of data is essential for training and validating our multi-task framework. Our strategy involved two key components: leveraging an established, high-quality benchmark for foundational kinetic tasks (k_{cat} , K_m), and constructing novel, rigorously partitioned datasets for the new tasks of catalytic efficiency and optimal environmental condition prediction (k_{cat}/K_m , pH, T_{opt}).

4.2. Experimental Setup

All models were trained on 8 NVIDIA A100 GPU using the AdamW optimizer (Loshchilov & Hutter, 2019). We employed a consistent training protocol with a maximum of 100 epochs, using an early stopping mechanism that monitored validation RMSE with a patience of 25 epochs to prevent overfitting. Recognizing the distinct statistical characteristics of each dataset, we optimized key hyperparameters such as learning rate and batch size on a per-task basis to ensure a fair and robust evaluation.

To analyze how different sources of information contribute to multi-property prediction. **Hybrid:** our base architecture, which encodes enzymes using ProtT5 and fuses enzyme-substrate representations through direct feature concatenation and a trainable cross-attention module. **Hybrid + ProstT5:** this variant replaces ProtT5 with ProstT5, enabling the model to incorporate structure-aware protein representations and quantify the effect of structural information. **Hybrid++:** on top of the Hybrid design, we further introduce auxiliary physicochemical descriptors: a 22-dimensional vector computed from the primary sequence using Biopython’s ProtParam, including amino acid composition, molecular weight, isoelectric point, and related features. Together, these three variants allow us to disen-

Table 1. Side-by-side performance comparison on the public CataPro benchmark datasets. Best results for each metric are highlighted in bold.

MODEL	k_{cat} DATASET			K_m DATASET		
	RMSE ↓	PCC ↑	SCC ↑	RMSE ↓	PCC ↑	SCC ↑
DLKCAT	1.426	0.368	0.349	1.052	0.578	0.578
UNIKP	1.382	0.448	0.449	1.039	0.599	0.594
ESM2_15B	1.431	0.435	0.467	1.063	0.581	0.567
ESM2_RDKITFP	1.380	0.435	0.424	1.025	0.609	0.607
ESM2_MORGAN	1.370	0.446	0.436	1.017	0.617	0.616
ESM2_MACC	1.356	0.463	0.453	1.007	0.625	0.622
ESM2_MOLT5	1.361	0.455	0.445	1.011	0.620	0.616
ESM2_MOLT5-MACC	1.357	0.464	0.459	1.004	0.627	0.623
PROT5_3B	1.441	0.443	0.434	1.068	0.603	0.519
PROT5_RDKITFP	1.389	0.421	0.415	1.029	0.604	0.602
PROT5_MORGAN	1.364	0.465	0.462	1.015	0.621	0.620
PROT5_MOLE-BERT	1.341	0.483	0.481	1.014	0.617	0.611
PROT5_MACC	1.338	0.486	0.482	1.011	0.621	0.616
PROT5_MOLT5	1.338	0.491	0.484	1.001	0.629	0.625
PROT5-SAPROT_MOLT5	1.323	0.490	0.489	1.004	0.626	0.620
CATAPRO	1.329	0.497	0.495	0.998	0.633	0.629
CATAPRO+SAPROT	1.317	0.493	0.489	0.999	0.631	0.625
ENZYME-UNIFIED (HYBRID)	1.304	0.498	0.484	0.996	0.643	0.628
ENZYME-UNIFIED (HYBRID+PROST5)	1.299	0.496	0.494	0.997	0.639	0.631
ENZYME-UNIFIED (HYBRID++)	1.301	0.495	0.419	0.999	0.643	0.624

tangle the contributions of sequence encoders, structural priors, and handcrafted physicochemical features. To ensure fair comparison, we first pair our Hybrid architecture with ProtT5 and benchmark it against ProtT5-based baselines, demonstrating that the architectural design itself provides clear advantages. We then present results from the ProstT5 variant, showing that the same architecture can further improve performance when equipped with a more advanced encoder, highlighting its scalability and future extensibility. The complete, task-specific hyperparameter configurations are detailed in Appendix A.

4.3. Main Results

We evaluated our ENZYME-UNIFIED hybrid architecture across all five predictive tasks, comparing its performance against a comprehensive suite of established baselines on both public benchmarks and our newly constructed datasets. The results consistently demonstrate the superiority of our hybrid approach, which not only establishes a new SOTA in kinetic prediction but also shows a dominant advantage in modeling complex environmental dependencies.

Our base hybrid model, without any structural enhancements, immediately establishes itself in the top tier of predictors. For k_{cat} prediction, it outperforms the vast majority of prior methods and achieves a new **state-of-the-art PCC of 0.498**. We then investigated whether incorporating explicit structural information via ProstT5 embeddings

could further boost performance. This enhanced model, hybrid+ProstT5, sets another **SOTA record with an RMSE of 1.299**.

Dominance on public kinetic benchmarks. To first validate the effectiveness of our architecture, we benchmarked it against 14 prior models on public, non-homologous CataPro datasets for catalytic rate (k_{cat}) and the Michaelis constant (K_m). The comprehensive results are presented in Table 1.

Compared to previous large-scale pretrained models (ProtT5-3B and ESM2-15B), our model also demonstrates strong performance on the k_{cat} : its RMSE of 1.299 represents a 9.2% relative improvement over the ESM2-15B baseline (1.431). The advantage is even more pronounced for K_m prediction. Our base hybrid model significantly outperforms all 14 baselines, including the strong CataPro models, on two of the three key metrics, establishing a new SOTA with an **RMSE of 0.996** and a **PCC of 0.643**. The hybrid+ProstT5 variant further pushes the performance boundary, achieving the **best SCC of 0.631**. These results confirm that our hybrid architecture, by balancing global and fine-grained features, provides a more powerful and extensible foundation for enzyme kinetic prediction.

Predicting catalytic efficiency Directly. While predicting k_{cat} and K_m separately is a standard benchmark, the direct prediction of catalytic efficiency (k_{cat}/K_m) is often more relevant for enzyme engineering and represents a more com-

Table 2. Comprehensive side-by-side performance comparison on our newly constructed datasets. For each model, we report performance across kinetics (k_{cat}/K_m), optimal pH, and optimal temperature (T_{opt}) prediction tasks. Each column’s best results are highlighted in bold.

MODEL	k_{cat}/K_m DATASET			OPTIMAL PH DATASET			OPTIMAL T_{opt} DATASET		
	RMSE ↓	PCC ↑	SCC ↑	RMSE ↓	PCC ↑	SCC ↑	RMSE ↓	PCC ↑	SCC ↑
CATAPro	1.716	0.368	0.366	2.167	0.521	0.427	17.553	0.661	0.495
UniKP	1.706	0.399	0.371	0.914	0.576	0.432	14.347	0.670	0.473
ProT5-3B	1.745	0.359	0.347	0.989	0.523	0.507	16.342	0.686	0.502
ESM2-15B	1.642	0.337	0.381	0.949	0.579	0.512	12.170	0.697	0.582
ENZYME-UNIFIED (HYBRID)	1.597	0.409	0.416	0.840	0.657	0.512	14.446	0.666	0.622
ENZYME-UNIFIED (HYBRID+PROST5)	1.601	0.403	0.421	0.901	0.613	0.510	12.433	0.703	0.612
ENZYME-UNIFIED (HYBRID++)	1.599	0.408	0.419	0.912	0.626	0.511	10.329	0.783	0.640

Table 3. Side-by-side ablation study on the key architectural components of ENZYME-UNIFIED. The model is evaluated across three distinct datasets: K_{cat} , K_m , and pH. Best results for each metric within each dataset are highlighted in bold.

Model Configuration	K_{cat} Dataset			K_m Dataset			pH Dataset		
	RMSE ↓	PCC ↑	SCC ↑	RMSE ↓	PCC ↑	SCC ↑	RMSE ↓	PCC ↑	SCC ↑
Concatenation-only	1.344	0.461	0.449	1.002	0.630	0.625	2.034	0.562	0.459
Attention-only	1.373	0.443	0.417	1.010	0.633	0.629	1.450	0.591	0.503
Hybrid w/ Simple Average	1.359	0.452	0.433	1.006	0.632	0.627	1.742	0.577	0.481
ENZYME-UNIFIED (hybrid)	1.304	0.498	0.484	0.997	0.639	0.631	0.840	0.657	0.512

plex learning task. (Fersht, 1977) We evaluated our framework on our newly constructed k_{cat}/K_m dataset, with results shown in Table 2.

On this challenging task, both variants of ENZYME-UNIFIED significantly outperform the strong UniKP and CataPro baselines. Our base hybrid model reduces the **RMSE by over 6.3%** compared to the next-best model (UniKP) and achieves the highest PCC of 0.409. The hybrid+ProstT5 variant demonstrates the best ranking correlation with an SCC of 0.421. This demonstrates our framework’s superior ability to learn the complex, non-linear relationship between sequence and catalytic efficiency on a challenging, rigorously partitioned dataset.

Superior performance on environmental property prediction. Our framework’s advantages are highly pronounced for environmental property prediction (Table 2). For optimal pH, our model achieves an RMSE of 0.840, a remarkable reduction of over 61% compared to the CataPro baseline, while increasing PCC by over 0.13. Compared to large-scale pretrained models such as ESM2-15B, our model’s PCC (0.657) is 13.5% better than the ESM2-15B baseline (0.579). On the challenging task of optimal temperature prediction, enhancing our model with amino acid physicochemical features (Kawashima et al., 2007; Gasteiger et al., 2005)(hybrid++) yields a dramatic performance leap, reducing the RMSE to an exceptional 10.329 and boosting the PCC to 0.783. Most strikingly, our model’s RMSE (10.329)

is 17.5% better than the ESM2-15B baseline (12.170) and 36.8% better than the ProT5-3B baseline (16.342). These results highlight our hybrid architecture’s effectiveness in modeling an enzyme’s environmental adaptation.

4.4. Ablation Study

To dissect the contributions of our key architectural components and feature choices, we conducted a series of ablation studies, with the core results presented in Table 3. The study validates three foundational aspects of our design.

First, the results unequivocally demonstrate the **synergy of the hybrid architecture**. Our full ENZYME-UNIFIED model substantially outperforms both the “Concatenation-only” and “Attention-only” pathways when used in isolation, confirming that the fine-grained pathway provides crucial complementary information to the global feature baseline. Second, the importance of the learnable **Gated Fusion mechanism** is validated, as it consistently yields better performance than a fixed averaging of the two pathways.

Finally, we confirm the framework’s ability to effectively leverage auxiliary information. While the benefit of some features can be task-dependent, incorporating amino acid physicochemical properties for T_{opt} **prediction led to a dramatic performance leap, reducing RMSE by nearly 41%** (detailed in Table 2). Collectively, these studies affirm that our hybrid design, adaptive fusion mechanism, and feature extensibility are essential for achieving state-of-the-

Table 4. Validating the fine-grained attention on RNase A’s catalytic histidines (His12, His119). Mutation of the critical H119A specifically disrupts the local attention pattern.

RNASE A VARIANT	REGION AT HIS12	REGION AT HIS119
WILD-TYPE (WT)	12-12	116-119
H119A (CATALYTIC MUTANT)	12-12	117-117
S16A (NEGATIVE CONTROL)	12-12	116-119

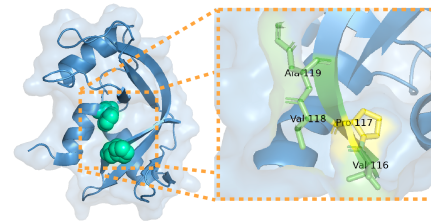


Figure 4. Interpretable attention on RNase A’s catalytic site. (Left) Wild-type RNase A. (Right) Mutating His119 causes attention collapse.

art performance. A more detailed discussion is provided in AppendixD.

4.5. Case Study: Interpreting the Attention Mechanism on Ribonuclease A

To validate that our model learns biochemically relevant principles, we conducted an in silico mutagenesis study on Ribonuclease A (RNase A) to probe the interpretability of our Fine-Grained Interaction Pathway (Rives et al., 2021). We hypothesized that the model’s attention should focus on the known catalytic histidines (His12, His119). (Crestfield et al., 1963). To test this, we compared the high-attention regions for the wild-type (WT) enzyme against a critical catalytic mutant (H119A) and a non-essential control mutant (S16A). As summarized in Table4, our findings confirm this hypothesis, providing strong evidence that the model learns to identify functional sites.

The Model Accurately Identifies Catalytic Histidines in WT RNase A. For the WT enzyme, the model’s high-attention regions precisely pinpointed the locations of the two catalytic histidines. As shown in Table4, the identified interaction regions include the segment 12-12 (His12) and the segment 116-119 (containing His119). This demonstrates that the model, without any explicit structural or functional labels, correctly learned to focus on the most important functional sites of the enzyme.

Mutation of a Key Histidine Specifically Disrupts Attention. Next, we assessed the effect of “knocking out” a critical residue (Smith, 1985; Wlodawer et al., 1988). Upon mutating the key catalytic histidine at position 119 to an alanine (H119A), the model’s attention pattern changed dramatically. The prominent, contiguous attention region spanning residues 116-119 in the WT enzyme was completely disrupted, collapsing to an isolated, single-residue focus at position 117-117. This strongly suggests that the model’s attention is not merely sequence-based but is directly correlated with the functional importance of the catalytic site (Vig et al., 2020).

The Negative Control Confirms Specificity. Finally, to confirm that this effect was specific to functional sites, we analyzed the S16A mutant. Serine 16 is not directly involved in catalysis (Raines, 1998). In this case, the mutation had a negligible impact on the attention patterns at the distant catalytic sites. The high-attention regions at 12-12 and 116-119 remained fully intact, identical to that of the WT enzyme. This result validates that the model’s response is highly specific and not a random artifact of sequence perturbation. For more examples, please refer to the Appendix F.

5. Conclusion

In this work, we introduced ENZYME-UNIFIED, a hybrid learning framework that addresses the fragmented, single-task paradigm by providing a holistic profile of an enzyme’s kinetic and environmental properties. Its novel hybrid architecture, validated by extensive ablation studies, synergistically combines global and fine-grained interaction pathways. Evaluated on three new, large-scale datasets with strict sequence-identity partitioning, our framework establishes a new state-of-the-art on kinetic benchmarks and shows dominant performance on environmental prediction. Crucially, a case study on RNase A confirmed the model’s biochemical interpretability, as its attention mechanism correctly identifies catalytic hotspots. ENZYME-UNIFIED thus provides a robust, extensible, and interpretable foundation for accelerating the design of novel biocatalysts.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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Appendix

A. Hyperparameter and Training Details

All models were trained on NVIDIA A100 GPUs. The core architecture for the Fine-Grained Interaction Pathway used a hidden dimension of 768 and 1 cross-attention layer. The specific hyperparameters used for each predictive task are detailed in Table 5.

Table 5. Task-specific hyperparameters for model training. All tasks used the AdamW optimizer and an early stopping patience of 25 epochs on the validation RMSE.

Predictive Task	Learning Rate	Batch Size
Kinetic (k_{cat})	1e-5	512
Kinetic (K_m)	1e-5	512
Kinetic (k_{cat}/K_m)	1e-5	512
Optimal Temperature (T_{opt})	1e-3	64
Optimal pH	5e-4	128

B. Details of Baseline Methods

In this section, we briefly describe the underlying principles of the key baseline and state-of-the-art methods used for comparison in our main experiments (Table 1).

B.1. Established Enzyme-Specific Frameworks

These are published models specifically designed for enzyme kinetic prediction.

- **DLKcat** is a pioneering deep learning framework developed for k_{cat} prediction. It utilizes a Convolutional Neural Network (CNN) to automatically extract features from a numerical representation of the protein’s primary sequence.
- **UniKP** is a multi-task framework designed to jointly predict both k_{cat} and K_m . Its architecture combines representations from a pre-trained protein language model (ESM-1v) for the enzyme with features from a Graph Neural Network (GNN) for the substrate molecule.
- **CataPro** represents a strong, modern baseline that also relies on the direct concatenation of features. It fuses global representations from a state-of-the-art protein language model (ESM-2) with molecular fingerprints derived from the substrate’s SMILES string. The CataPro+SaProt variant further enhances the protein representation by incorporating features from SaProt, a structure-aware protein model.

B.2. General-Purpose Encoder Baselines

To establish a comprehensive and fair comparison, we constructed a suite of strong baseline models by combining state-of-the-art protein and molecule encoders. This allows us to assess the performance of a standard ”concatenation-only” approach using the best available off-the-shelf components.

Protein Encoders We used two powerful, pre-trained protein language models:

- **ESM-2** is a transformer-based protein language model trained exclusively on tens of millions of protein sequences. It excels at generating context-aware embeddings that capture evolutionary and biophysical information.
- **ProtT5** is a sequence-to-sequence language model trained on a similar scale to ESM-2. Its architecture is particularly effective at capturing functional features from protein sequences. We also test a +SaProt variant, which enriches the sequence representation with structural information.
- **ProstT5** (Protein structure-sequence T5) is a protein language model (pLM) which can translate between protein sequence and structure. It is based on ProtT5-XL-U50, a T5 model trained on encoding protein sequences using span corruption applied on billions of protein sequences. ProstT5 finetunes ProtT5-XL-U50 on translating between protein sequence and structure using 17M proteins with high-quality 3D structure predictions from the AlphaFoldDB. Protein structure is converted from 3D to 1D using the 3Di-tokens introduced by Foldseek.

Substrate Encoders The global protein embeddings were concatenated with various substrate representations to form the final input for the prediction head:

- **RDKitFP, Morgan, MACC:** These are widely-used, standard molecular fingerprints. They are fixed-length bit vectors calculated from the substrate’s 2D structure (SMILES string) and represent the presence or absence of specific chemical substructures.
- **MolT5 / Mole-BERT:** These are powerful, pre-trained transformer models for small molecules. Unlike static fingerprints, they generate context-aware embeddings for the substrate, capturing deeper chemical and semantic information.

The performance of these various combinations (e.g., ESM2_MACC, ProtT5_MolT5) represents a robust but standard approach where global features from both modalities are simply concatenated before prediction.

B.3. Fine-Grained Interaction Pathway

The Fine-Grained Interaction Pathway in ENZYME-UNIFIED relies on several implementation components that enable robust token-level modeling of enzyme-substrate interactions. Since enzyme sequences and substrate SMILES strings differ substantially in length, all sequences are padded within each batch to the maximum modality-specific length. To ensure that padded positions never influence attention computation, we apply an attention mask during multi-head cross-attention: all logits corresponding to padded tokens are assigned large negative values before the softmax operation, effectively excluding them from the attention distribution. This mechanism allows the architecture to handle variable-length inputs reliably during batched training.

Importantly, the cross-attention module does not depend on any predefined structural or biochemical alignment between enzyme residues and substrate atoms. Instead, it learns these correspondences dynamically. Each substrate token attends over all enzyme tokens, and each enzyme token symmetrically attends over all substrate tokens, forming a fully connected interaction graph. Through end-to-end optimization, the model identifies the residues most relevant to each substrate fragment, enabling it to capture catalytic signals, binding preferences, and other fine-grained biophysical relationships directly from data.

To supply positional context for enzyme and substrate sequences, the pathway leverages the absolute positional embeddings already learned by the pre-trained ProtT5 and MolT5 encoders. These embeddings provide robust information about sequential order and local neighborhoods, supporting the discovery of fine-grained interactions across long protein sequences. Although relative positional embeddings are not used in the current implementation, they represent a natural extension for future refinement and may offer benefits for modeling long-range residue dependencies.

In combination, padding with masked attention, dynamic cross-attention without predefined alignments, and pre-trained absolute positional embeddings enable the Fine-Grained Interaction Pathway to model enzyme-substrate interactions at high resolution while remaining architecture-agnostic and fully end-to-end.

C. TRAINING PARADIGMS AND CROSS-TASK SYNERGY

C.1. Training Paradigms

To further clarify the role of the unified architecture and address the question of how different tasks interact within ENZYME-UNIFIED, we conduct two additional analyses: (1) a direct comparison between specialized and true multi-

task training using a shared hybrid trunk, and (2) a transfer-learning-based evaluation of cross-task synergy. These analyses complement the main results and provide a deeper understanding of how the unified backbone supports multi-property modeling.

Although ENZYME-UNIFIED is designed as a unified architectural framework, the optimal training paradigm may depend on dataset size, noise, and task heterogeneity. To evaluate this, we compare three strategies under identical training conditions: **Specialized models**: where ENZYME-UNIFIED is trained separately for each task; **Multi-task ENZYME-UNIFIED**: where a shared hybrid trunk feeds five task-specific heads trained jointly; **Multi-task CataPro**: where the previous SOTA is adapted to the same multi-task setting. The results are shown in the Table 6.

Across all five prediction tasks, the unified architecture consistently outperforms CataPro under the same multi-task regime, confirming the advantage of our hybrid design independent of training strategy. Meanwhile, specialized training yields the best overall performance on current homology-filtered datasets (e.g., Topt RMSE 10.33 vs 14.51 vs 18.77 for the three strategies above). This indicates that while the architecture is unified, specialized optimization remains empirically favorable on existing benchmarks, likely due to dataset scale and inter-task imbalance.

C.2. Cross-Task Synergy

To examine whether the unified architecture captures transferable biochemical features across properties, we conduct an explicit cross-task transfer experiment. We pretrain ENZYME-UNIFIED on optimal pH prediction, freeze the encoder, and fine-tune a new regression head on Topt. This “pH $\rightarrow$ Topt” transfer setting isolates representational sharing from optimization-level interactions.

The transfer model achieves substantial improvements over training on Topt alone, reducing RMSE from 14.17 to 11.11 (-21.6%) and increasing PCC from 0.617 to 0.706. This demonstrates that the model learns biophysically meaningful and task-general representations e.g., sequence determinants of stability, that benefit multiple properties. The results are shown in the Table 7.

These findings empirically support our holistic motivation: although tasks are optimized independently in the main results for performance reasons, ENZYME-UNIFIED does learn cross-property biochemical signals that are transferable across tasks.

D. Detailed Ablation Study Analysis

This section provides a more detailed discussion of the ablation studies summarized in Section 4.4 and Table 3 of

Table 6. Head-to-head comparison of training paradigms. ENZYME-UNIFIED (Ours) and CataPro are evaluated under both Specialized (one model per task) and Multi-Task (shared trunk, task-specific heads) settings. Best performance within each paradigm is in **bold**.

Task	Metric	Specialized		Multi-Task	
		Ours	CataPro	Ours	CataPro
k_{cat}	RMSE ↓	1.304	1.329	1.306	1.339
	PCC ↑	0.498	0.497	0.493	0.483
	SCC ↑	0.484	0.495	0.495	0.481
K_m	RMSE ↓	0.996	0.998	0.998	1.009
	PCC ↑	0.643	0.633	0.635	0.629
	SCC ↑	0.631	0.629	0.627	0.626
k_{cat}/K_m	RMSE ↓	1.597	1.716	1.664	1.764
	PCC ↑	0.409	0.368	0.405	0.358
	SCC ↑	0.416	0.366	0.395	0.360
Optimal pH	RMSE ↓	0.840	2.167	0.923	1.223
	PCC ↑	0.657	0.521	0.584	0.513
	SCC ↑	0.512	0.427	0.473	0.433
Optimal T_{opt}	RMSE ↓	10.329	17.553	14.508	18.768
	PCC ↑	0.783	0.661	0.717	0.617
	SCC ↑	0.640	0.495	0.616	0.436

Table 7. Direct evidence for cross-task synergy on the Optimal T_{opt} prediction task. This table compares the performance of our hybrid model trained via transfer learning against all standard baselines and our fully trained models. The ‘Transfer’ model, pretrained on Optimal pH data, shows a dramatic performance improvement over the ‘Baseline’ trained from scratch, demonstrating that our hybrid model learns transferable biophysical representations.

Model Setting	RMSE on T_{opt} ↓	PCC on T_{opt} ↑	SCC on T_{opt} ↑
<i>General Baselines</i>			
CataPro	17.553	0.661	0.495
UniKP	14.347	0.670	0.473
ProtT5-3B	16.342	0.686	0.502
ESM2-15B	12.170	0.697	0.582
<i>Our Models (Specialized Training)</i>			
ENZYME-UNIFIED (hybrid)	14.446	0.666	0.622
ENZYME-UNIFIED (hybrid+ProtT5)	12.433	0.703	0.612
ENZYME-UNIFIED (hybrid++)	10.329	0.783	0.640
<i>Synergy Experiment: Transfer vs. From Scratch</i>			
Baseline (Train on T_{opt} from scratch)	14.17	0.617	0.607
Transfer (Pre-train on pH → Fine-tune on T_{opt})	11.11	0.706	0.640

hybrid model’s performance surpasses even this strong baseline. This indicates that while the “Attention-only” pathway is less effective in isolation, it captures distinct, complementary features that are vital for refining the model’s predictions. The synergy of combining both pathways is therefore essential.

D.2. The Importance of the Gated Fusion Mechanism

We investigated the importance of our trainable gating parameter, α , by comparing our standard model against a variant using a simple, fixed average (i.e., $\sigma(\alpha)$ is fixed to 0.5). As shown in Table 3, the learnable gate provides a consistent performance improvement. This suggests that allowing the model to learn the optimal, task-specific balance between the global and fine-grained pathways is a beneficial architectural refinement that contributes to the final performance.

D.3. Quantifying the Impact of Enhanced Features

Here we provide a more detailed analysis of the performance gains from incorporating advanced features, referencing results from our main experiments (Table 2).

Kinetic Prediction Incorporating structural embeddings from ProtT5 yielded only a marginal improvement. We hypothesize that this is due to a high degree of informational redundancy for this task. With over 90% of the enzymes in our training set having structures predicted by AlphaFold2,

the main text.

D.1. The Synergy of the Hybrid Architecture

Our central hypothesis is that a hybrid model combining global and fine-grained features outperforms its constituent pathways. To test this, we trained two ablated models: (1) a **Concatenation-only** model and (2) an **Attention-only** model. As shown in Table 3, the full hybrid model substantially outperforms both across all metrics. Between the individual pathways, the “Concatenation-only” model establishes a stronger baseline. Crucially, however, the full

the 3D tokens from ProstT5 may exhibit strong collinearity with the primary sequence features they are derived from, limiting the amount of novel, independent information the model can extract. It is possible that the global structural representations from ProstT5, while informative, lack the fine-grained resolution to capture the subtle conformational dynamics crucial for kinetics.

Environmental Prediction In stark contrast, the impact was significantly more pronounced for environmental prediction. Supplementing our model with amino acid physicochemical features for T_{opt} prediction resulted in a dramatic performance leap, **reducing the RMSE by nearly 41%** (from 17.55 to 10.33) and **increasing the PCC by over 0.12**. We attribute this substantial improvement to the fact that these features, such as the aliphatic index, provide crucial biophysical information directly related to thermal stability—information that is not explicitly or easily derivable from sequence or global structure alone. This suggests they offer a new, orthogonal source of information that is highly relevant for modeling an enzyme’s adaptation to temperature.

E. DATA PROCESSING

To guarantee an unbiased and generalizable evaluation, we constructed our dataset using a rigorous clustering-based splitting protocol: all enzyme sequences were first clustered with MMseqs2 at a 40 % identity threshold, and then entire clusters—rather than individual samples—were distributed into non-overlapping training, validation and test folds. This ensures that the maximum sequence similarity between any enzyme in different folds never exceeds 40%, forcing the model to generalize across protein families. Substrate or assay meta-information were retained in their natural distribution across folds, were not used as splitting criteria, and received no additional annotation, eliminating any implicit leakage. The procedure follows best practices in protein learning, and Table 8 is a detailed description of each dataset. Consequently, our dataset has been strictly processed and is free of data-leakage issues.

Table 8. The k_{cat}/K_m data contain 10 folds; the first fold comprises 2,637 samples, and each of the remaining nine folds contains 2,636 samples. The pH dataset consists of 5 folds, each with 104 samples. The temp dataset has 5 folds; the first fold includes 142 samples, and the other four folds each contain 141 samples.

Dataset	Total Samples	Seq. Id. Threshold	Clusters	Folds	Samples per Fold Distribution
k_{cat}/K_m	26,361	0.4	3329	10	All Folds 1: 2,637; Folds 2–10: 2,636
pH	520	0.4	205	5	All Folds: 104
Temperature	706	0.4	251	5	Fold 1: 142; Folds 2–5: 141

F. CASE STUDY

This appendix provides additional interpretability experiments to evaluate the robustness and generality of the ENZYME-UNIFIED cross-attention mechanism across different enzyme classes, catalytic mechanisms, and sequence scales. We selected three representative enzymes: **TEM-1 Beta-Lactamase**, **Alcohol Dehydrogenase (ADH)**, and **Beta-galactosidase (LacZ)**. All analyses use raw cross-attention weights without any structural labels or task-specific fine-tuning.

TEM-1 Beta-Lactamase (EC 3.5.2.6): The model consistently highlights residues 65–70, including the catalytic nucleophile Ser70. This indicates that ENZYME-UNIFIED can identify the beta-lactam hydrolysis motif directly from sequence. **Alcohol Dehydrogenase (ADH, EC 1.1.1.1)**: For this zinc-dependent oxidoreductase, the model focuses on residues 70–72 and 169–172, aligning with the catalytic zinc-binding residues His67 and Cys174. **Beta-Galactosidase (LacZ, EC 3.2.1.23)**: For this large (>1000 aa) enzyme, ENZYME-UNIFIED highlights regions 460–462 and 525–538, capturing the key catalytic residues Glu461 and Glu537.

Across these diverse enzymes, ENZYME-UNIFIED reliably identifies catalytic centers, demonstrating generalizable interpretability beyond RNase A.

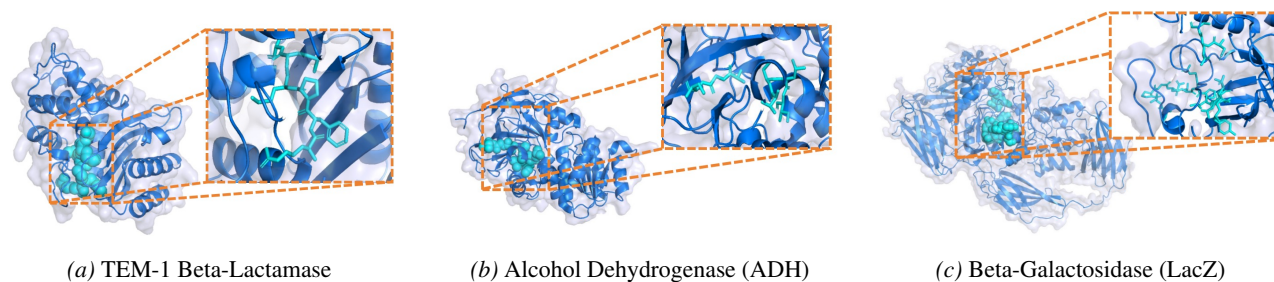


Figure 5. Cross-attention visualizations for three representative enzymes across different EC classes. ENZYME-UNIFIED consistently highlights known catalytic residues without structural supervision.